Customer Churn & Revenue Risk Analysis

A Data-Driven Approach to Protecting Telecom Revenue

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The Business Problem

Telecom companies face high customer churn rates that directly threaten recurring revenue streams and long-term profitability. Understanding and addressing churn is critical to revenue protection.

Strategic Objectives

- Identify and validate primary churn drivers across customer segments
- Quantify monthly and annual revenue at risk from customer attrition
- Prioritize retention strategies based on financial impact and feasibility

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Dataset Overview

7,043

Customers

Total records analyzed

21

Features

Data attributes examined

4

Data Categories

Demographics, contracts, billing, services

The dataset includes comprehensive customer information with demographics, contract terms, billing details, and service usage patterns. The target variable is churn status, enabling predictive analysis of attrition patterns.

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Data Preparation Process

Data Cleaning

Removed duplicates and corrected inconsistencies

Missing Values

Imputed or removed incomplete records

Feature Engineering

Created new predictive variables

Tenure Segments

Binned into lifecycle stages

Comprehensive data preparation ensured data quality and created meaningful customer segments for analysis. Tenure was categorized into 0-12 months (early lifecycle), 13-24 months (mid-cycle), and 25+ months (established customers) to identify lifecycle-specific churn patterns.

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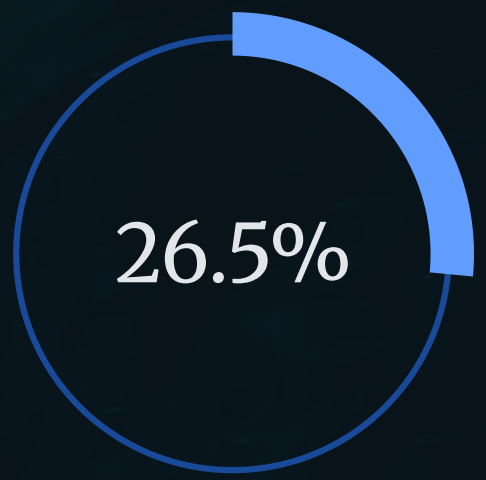
Churn Overview: Critical Metrics



Churn Rate

Customers leaving monthly

Revenue Impact

Share of total revenue lost

Monthly Revenue Streams

- Total monthly revenue: **$456,000**
- Monthly churn revenue loss: **$139,000**
- Annualized revenue at risk: **$1.67M**

Strategic Insight

Churned customers represent 30.5% of total revenue loss, indicating that high-value customers are disproportionately at risk. This concentration suggests targeted retention efforts could yield significant ROI.

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Key Churn Drivers

Contract Type

Month-to-month contracts show 3.2× higher churn vs. 1-2 year contracts

Customer Tenure

Early lifecycle customers (0-12 months) have 2.8× higher churn risk

Monthly Charges

Customers paying above median ($70/month) have 1.9× higher churn probability

Statistical analysis revealed contract type, tenure, and monthly charges as the top three predictors of customer attrition. Each driver presents distinct retention opportunities.

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Highest Risk Segment

1

Segment Definition

Month-to-month contracts with tenure of 0-12 months

2

Churn Rate: 51.4%

Over half of this segment churns monthly

3

Revenue Impact: ~$68K

Monthly revenue lost from this segment alone

This high-risk segment represents the greatest opportunity for immediate retention impact. Strategic interventions targeting this group could recover approximately $816K annually in lost revenue.

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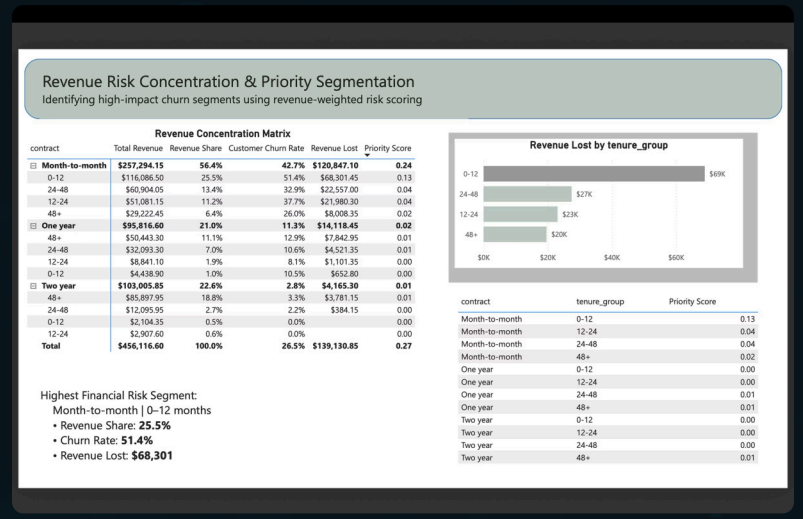
Dashboard Insights



Revenue Risk Concentration

Visual heatmaps showing revenue loss by contract type and tenure segment

Customer Segmentation

Interactive filters to explore churn patterns across demographics and services

KPI Monitoring

Real-time tracking of churn rate, revenue at risk, and retention program effectiveness

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Statistical Validation

01

Chi-Square Test

Contract type vs. churn status: $\chi^2(2) = 1{,}247$, $p < 0.001$

02

T-Test (Monthly Charges)

Mean difference: $18.50 higher for churned, $t(7{,}041) = 15.2$, $p < 0.001$

03

T-Test (Tenure)

Mean difference: 19.2 months shorter for churned, $t(7{,}041) = -22.8$, $p < 0.001$

All three churn drivers demonstrated **statistical significance at $p < 0.001$**, confirming they are genuine predictors of customer attrition rather than random variation. Effect sizes were medium to large, supporting practical significance for business interventions.

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Business Recommendations

Focus Segments

- Early lifecycle customers (0-12 months)
- Month-to-month contract holders
- High monthly charge customers (> $70)

Contract Conversion

Launch targeted campaigns offering 6-12 month contract incentives with $5-10/month discounts

Onboarding Programs

Deploy proactive engagement for first 90 days including welcome calls and usage education

Value Optimization

Review pricing tiers for customers above $70/month; consider value-add bundles or loyalty discounts

Implementation of these strategies could reduce churn by 8-12 percentage points, recovering $350K-500K in annual recurring revenue. Pilot programs should target the highest-risk segment first for maximum ROI.

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